Velo tracking efficiencies in 2025 pre-TS

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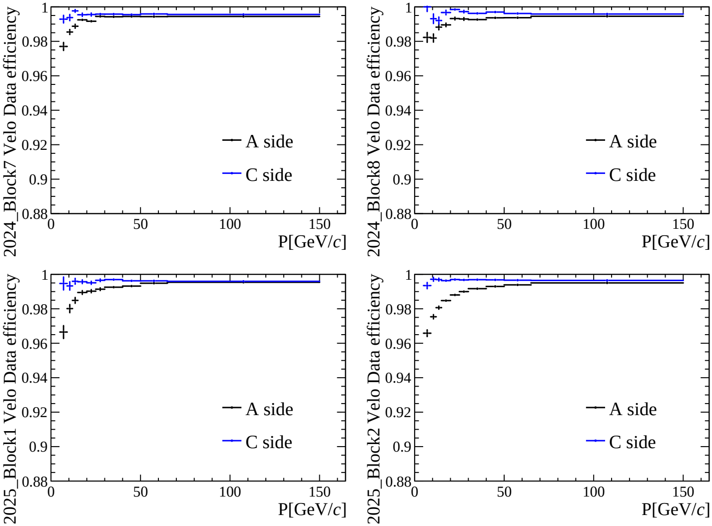
Introduction

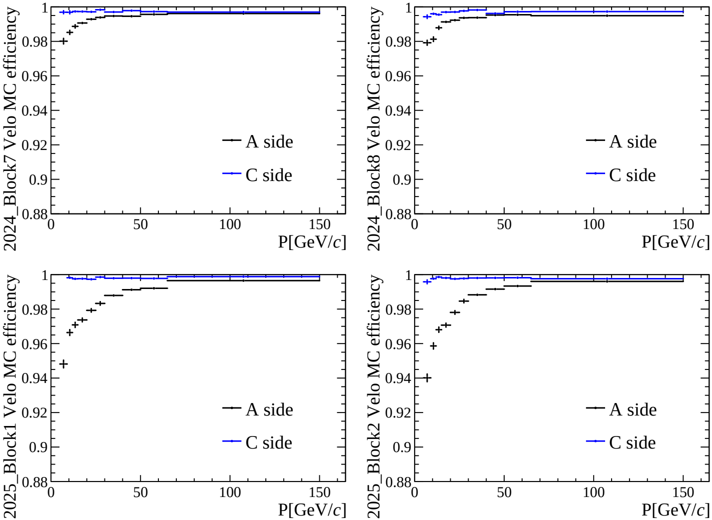
Tracking efficiencies presented at A&S week:

- 2024 tracking efficiencies are available to analysts here
  - Version 0: Smaller MC samples and with only the main systematic
  - Next release: Updated 2D binning scheme, larger MC samples, multiplicity reweighting
- 2025 efficiencies are currently not released
  - Velo efficiencies had a large discrepancy between data and MC
  - Unusual behaviour: MC significantly underestimates the tracking efficiency
  - Feedback at A&S week: Check left-right asymmetries and check if M21/M23 are turned off in MC

Velo left-right asymmetry



Velo left-right asymmetry



Velo M21 and M23

Tracking efficiencies presented at A&S week:

- M21 and M23 sit at $z = 6.875\,\mathrm{cm}$ and $9.375\,\mathrm{cm}$, respectively
- Each module has 4 sensors
- M23 in data taking:
  - 2024: All sensors excluded
  - 2025 pre-TS: All sensors working
  - 2025 post-TS: 2 sensors working
- M23 in MC:
  - 2024: All sensors excluded
  - 2025 pre-TS: All sensors excluded
  - 2025 post-TS: 2 sensors working
- Furthermore, 2 sensors in M21 were not working in 2025 and also turned off in MC
  - This module is directly in front of M23, so could have significant impact on Velo tracking efficiencies!

Conclusion and next steps

We know 2025 pre-TS MC is wrong, what now...?

- Simulation group knows and MR with fix was opened a month ago...
- ... but this is low priority and will not be fixed until Sim10i
  - We're currently on Sim10g!
- Plan: We need to provide Sim10g corrections to analysts, but be prepared to produce a new set of corrections once Sim10i is available
- From my side, apart from this data/MC discrepancy, I am ready to release tracking efficiency correction tables for 2025 pre-TS
  - Need to request MC for the rest of 2025
  - Should in principle have a better data/MC agreement